

# Tianyu Kong

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**Website:** timkong98.github.io

**Research interests** Mathematical modeling, numerical analysis, partial differential equations, mathematical physics, quantum physics, and optimization

**Education** **University of Minnesota Twin Cities** Minneapolis, MN  
PhD in Applied Mathematics 08.2021 – 2026 (Expected)  
Advisor: Mitchell Luskin, Alexander Watson  
GPA: 3.96.

**University of Chicago** Chicago, IL  
Honors BS in Applied Mathematics & Physics 09.2017 – 06.2021  
GPA: 3.73.

**Honors and scholarships** SIAM Student Travel Award 03.2024  
Vanky Men Memorial Fellowship (UMN) 12.2023  
Dean's List (UChicago) 07.2018, 07.2019

**Publications** **Modeling of electronic dynamics in twisted bilayer graphene**  
T Kong, D Liu, M Luskin, AB Watson  
*SIAM J. Appl. Math.* 84, 1011 (2024).

**Bistritzer-MacDonald dynamics in twisted bilayer graphene**  
AB Watson, T Kong, AH MacDonald, M Luskin  
*Journal of Mathematical Physics* 64.3 (2023). *Editor's Choice*

**Research experience** **Graduate Research Assistant, UMN**  
Advisor: Mitchell Luskin and Alexander Watson 06.2021 – Present  
My research is on mathematical modeling and numerical analysis for incommensurate 2D materials, specifically twisted bilayer graphene. The main focus of my research is accurately computing the band structure and single electron dynamics in tight binding models of such systems. I use multi-scale analysis to identify a regime where the discrete aperiodic model can be reduced to a periodic continuum model. I also derive a domain truncation scheme to approximate the electron dynamics in an infinite system. I implement Python code to numerically verify my findings.

### **Student Assistant, Lawrence Berkeley National Laboratory**

Advisor: Chao Yang and Lin Lin

05.2024 – 08.2024

05.2023 – 08.2023

I studied exotic quantum phases in magic angle twisted bilayer graphene using interacting models that accounts for electron-electron correlation. I implemented methods including Hartree-Fock and coupled cluster theory with Python-based Simulation of Chemistry Framework (PySCF). I developed an accurate continuum model to compute the electron dispersion, and computed long range Coulomb interaction between electrons. I performed various numerical experiments, and tried to identify the phase of the ground states of such systems.

### **Undergraduate Research Assistant, UChicago**

Advisor: Mary Silber

04.2020 – 05.2021

I studied different types of Minimum Action Methods to determine the transition of states in a dynamical system subject to random perturbation. I also implemented MATLAB code and used steepest descent and quasi-Newton methods to calculate the Minimum Action Path in Lorenz systems.

### Teaching experience

#### **Teaching assistant, School of Mathematics, UMN**

09.2021 – 05.2024

*Honors Calculus*

*Calculus for College of Science and Engineering*

I held 2 workshop sessions per week for 3 hours total. In these sessions I lectured and organized group discussions on challenging concepts and problems, and responded to students' questions in an engaging manner. I also utilized various activities in class to enhance their understanding and spark their interest. I graded weekly homework, quizzes and final exams promptly, and provided valuable feedback.

#### **Grader, School of Mathematics, UMN**

09.2022 – 05.2024

*Mathematical Modeling and Applied Mathematics*

*Functional Analysis*

Promptly graded weekly homework and exams for  $\sim 30$  students.

#### **Grader, Department of Mathematics, UChicago**

09.2018 – 05.2020

*Math Methods for Physical Sciences*

*Honors Calculus*

Promptly graded weekly homework and exams for  $\sim 30$  students.

### Conference presentations

#### **Modeling of Electronic Dynamics in Twisted Bilayer Graphene** 07.2024

International workshop on 2D and moiré materials, Roscoff France

#### **Modeling of Electronic Dynamics in Twisted Bilayer Graphene** 05.2024

SIAM Conference on Materials Science (MS24), Pittsburgh

**Modeling of Electronic Dynamics in Twisted Bilayer Graphene** 03.2024  
Brin Mathematics Research Center Workshop, College Park

**A Comparison of Minimum Action Methods for Computing Noise-induced Transitions of the Lorenz System** 05.2021  
Poster, SIAM Conference on Applications of Dynamical Systems (DS21)

**Skills**

**Programming**

Python, MATLAB, Julia, R, MySQL

**Professional memberships**

**SIAM student chapter, UMN**

07.2023 – Present